LINKED LISTS & TREES Meta

COMPUTER SCIENCE MENTORS 61A

October 5, 2026 – October 9, 2026

Example Timeline

- **This is a menu of problems, not a requirement to finish the worksheet.** Individual suggested times do not add up to one hour. Choose or skip problems based on your students, and give each mini-lecture immediately before its section.
- Check-in and icebreaker [5 min].
- Linked lists: Q1 or Q2 [15 min]. Choose Q1 for combining lists with `Link` (append, then reverse) or Q2 for filtering an existing list. Q3 (merging sorted lists) is extra practice with multiple base cases.
- Trees: Q4, then Q5 [15 min]. Have students draw the tree before reading the code.
- Tree processing: Q6 or Q7 [20 min]. Choose Q6 for returning a list of linked lists or Q7 for passing a shrinking target down the tree.
- Wrap-up and questions [5 min]. Ask students to compare how a linked list and a tree are each defined recursively.

This Week

- **Midterm 2 is this week (10/5–10/9)** and covers material through linked lists. Students may want more linked-list review than tree practice, so consider spending extra time on Q1–Q3. You may also run section as office hours this week.
- **HW4 is due Thursday 10/8.** HW4 Q2 `interleave` is close to Q3 (`merge`), and HW4 Q5 `max_path_sum` is close to Q5 (`tree_height`). Guide students through the worksheet versions rather than solving the homework directly.

Teaching Goals

- A linked list is either empty (`()`) or a `first` value and a `rest` that is a linked list. Check **`isinstance`**`(s, Link)` before using `first` or `rest`.
- A tree is a `label` and a list of `branches`, each a tree. Trees cannot be empty, so a leaf is the usual base case.
- For both structures, use the recursive leap of faith: say what the recursive call returns, then decide how to combine it with the current node.
- Every second argument to `Link` must be a linked list, and every `branches` value must be a list of trees. Have students name the type of each recursive result before writing the combine step.
- These problems read and construct values; they do not mutate them.

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1 Linked Lists

For the following questions, use this dataclass-based linked list. A `LinkedList[T]` is either a `Link[T]` or the empty tuple `()`. The type variable `T` describes the element type.

```
from __future__ import annotations
from dataclasses import dataclass

type LinkedList[T] = Link[T] | tuple[()]

@dataclass
class Link[T]:
    first: T
    rest: LinkedList[T] = ()

    def __str__(self) -> str:
        return format_link(self)

def format_link[T](s: Link[T]) -> str:
    string = '(' + str(s.first)
    remaining = s.rest
    while isinstance(remaining, Link):
        string += ' ' + str(remaining.first)
        remaining = remaining.rest
    assert remaining == (), 'Not a valid linked list'
    return string + ')'
```

Construct a list with `Link(1, Link(2))`. Read its elements with `first` and `rest`. At the end, `rest` is `()`, which has no `first` or `rest` attributes. **`print`** uses parentheses around the elements; entering a `Link` at the prompt shows the dataclass representation with field names.

Teaching the interface: Establish construction, field access, and the empty case first. Treat `format_link` as supplied display behavior; students need not implement it. In the questions, reason about element values and shorter lists. If using diagrams, draw the sequence structure without asking students to distinguish shared nodes from equal-valued nodes.

This definition uses Python 3.12 or later. The future import also supports these annotations on Python 3.12–3.13. Reference: Composing Programs, Compound Data.

1. Write `append`, which takes two linked lists and returns a new linked list with all of the elements of the first followed by all of the elements of the second. Then use `append` to write `reverse`, which returns a new linked list with the elements in reverse order.

```
def append[T](a: LinkedList[T], b: LinkedList[T]) -> LinkedList[T]:
    """
    >>> print(append(Link(1, Link(2)), Link(7, Link(8))))
    (1 2 7 8)
    >>> print(append((), Link(5)))
    (5)
    >>> append((), ())
    ()
    """
    if ____________________________________________:

        return ____________________________________

    return ____________________________________________________________


def reverse[T](lnk: LinkedList[T]) -> LinkedList[T]:
    """
    >>> print(reverse(Link(1, Link(2, Link(3)))))
    (3 2 1)
    >>> reverse(())
    ()
    """
    if ____________________________________________:

        return ____________________________________

    return ____________________________________________________________
```

```
def append[T](a: LinkedList[T], b: LinkedList[T]) -> LinkedList[T]:
    if not isinstance(a, Link):
        return b
    return Link(a.first, append(a.rest, b))

def reverse[T](lnk: LinkedList[T]) -> LinkedList[T]:
    if not isinstance(lnk, Link):
        return ()
    return append(reverse(lnk.rest), Link(lnk.first))
```

Teaching Tips

Suggested Time: 10 min; Difficulty: Easy–Medium

- **Append:** When `a` is empty, the answer is just `b`. Otherwise keep `a.first` on the front and append `b` to the rest of `a`.
- **Reverse:** Assume `reverse(lnk.rest)` returns the rest reversed. The first element belongs at the end, so it is appended as a one-element list.

- **Types:** `Link(lnk.first)` is a one-element linked list, so it is a valid second argument to `append`.

2. Write `keep_if`, which takes a linked list of integers `lnk` and a predicate `pred`, and returns a new linked list of only the elements for which `pred` returns true, in their original order.

```
from collections.abc import Callable
def keep_if(
    lnk: LinkedList[int], pred: Callable[[int], bool]
) -> LinkedList[int]:
    """
    >>> lnk = Link(1, Link(2, Link(3, Link(4))))
    >>> print(keep_if(lnk, lambda x: x % 2 == 0))
    (2 4)
    >>> print(keep_if(lnk, lambda x: x > 0))
    (1 2 3 4)
    >>> keep_if(lnk, lambda x: x > 10)
    ()
    """
    if _____________________________________________:

        return _____________________________________

    rest = _____________________________________________

    if _____________________________________________:

        return _____________________________________

    return _____________________________________________
```

```
def keep_if(
    lnk: LinkedList[int], pred: Callable[[int], bool]
) -> LinkedList[int]:
    if not isinstance(lnk, Link):
        return ()
    rest = keep_if(lnk.rest, pred)
    if pred(lnk.first):
        return Link(lnk.first, rest)
    return rest
```

Teaching Tips

Suggested Time: 8 min; Difficulty: Easy–Medium

- **Leap of faith:** `keep_if(lnk.rest, pred)` is the filtered version of the rest. Only the first element is left to decide.
- **Two outcomes:** Keep `first` (put it on the front of the filtered rest) or drop it (return the filtered rest unchanged).
- **Connect to lists:** This is the linked-list version of a list comprehension with an **`if`** clause.

3. Write `merge`, which takes two linked lists of integers, each sorted in increasing order, and returns a new sorted linked list containing all of the elements of both.

```
def merge(a: LinkedList[int], b: LinkedList[int]) -> LinkedList[int]:
    """
    >>> a = Link(1, Link(5))
    >>> b = Link(2, Link(3, Link(8)))
    >>> print(merge(a, b))
    (1 2 3 5 8)
    >>> print(merge(a, ()))
    (1 5)
    >>> merge((), ())
    ()
    """
    if _____________________________________________:

        return _____________________________________

    elif _____________________________________________:

        return _____________________________________

    elif _____________________________________________:

        return _____________________________________

    else:

        return _____________________________________
```

```
def merge(a: LinkedList[int], b: LinkedList[int]) -> LinkedList[int]:
    if not isinstance(a, Link):
        return b
    elif not isinstance(b, Link):
        return a
    elif a.first <= b.first:
        return Link(a.first, merge(a.rest, b))
    else:
        return Link(b.first, merge(a, b.rest))
```

Teaching Tips

Suggested Time: 10 min; Difficulty: Medium

- **Base cases:** If either list is empty, the answer is the other list (which may also be empty).
- **Smallest first:** The smallest remaining element is `a.first` or `b.first`. Put it on the front, then merge the remaining elements.
- **Which arguments shrink:** Only the list whose first element was used gets shorter; the other is passed unchanged.

2 Trees

For the following question, use this dataclass-based tree. A `Tree[T]` has a `label` of type `T` and a list of `branches`, each of which is also a `Tree[T]`. A tree is never empty.

```
from __future__ import annotations
from dataclasses import dataclass, field

@dataclass
class Tree[T]:
    """A Tree has a label (of type T) and a list of branches,
    which are trees.

    >>> t = Tree(3, [Tree(1, []), Tree(2, [Tree(1, []), Tree(1, [])])])
    >>> t.branches[0]
    Tree(label=1, branches=[])
    >>> len(t.branches)
    2
    >>> t.branches[1].label
    2
    >>> len(t.branches[1].branches)
    2
    >>> expr = Tree(10, [Tree(6, [Tree(2), Tree('*'), Tree(3)]),
    ...                  Tree('+'), Tree(4)])
    >>> print(expr)
    10
      6
        2
        *
        3
      +
      4
    """
    label: T
    branches: list[Tree[T]] = field(default_factory=list)  # branches
       defaults to []

    def __str__(self):
        return format_tree(self)

def format_tree(t: Tree, indent='') -> str:
    "Format a tree with each branch indented below its label."
    assert isinstance(t, Tree), f'{t!r} is not a Tree'
    assert isinstance(t.branches, list), f'branches of {t!r} is not a list'
    string = indent + str(t.label)
    for b in t.branches:
        string += '\n' + format_tree(b, indent + '  ')
    return string

def is_leaf(t: Tree) -> bool:
    "Return whether a Tree t is a leaf with no branches."
    return not t.branches
```

4. What would Python display? Run these statements in order. If a statement causes an error, identify the error and continue.

```
>>> t = Tree(5, [Tree(3, [Tree(8)]), Tree(1), Tree(7)])
>>> t.label

5

>>> t.branches[0].label

3

>>> is_leaf(t.branches[1])

True

>>> len(t.branches)

3

>>> t.branches[1].branches[0]

IndexError: list index out of range

>>> t.branches[0].branches

[Tree(label=8, branches=[])]

>>> u = Tree(2, [t.branches[0]])
>>> u.branches[0].branches[0].label

8

>>> print(u)

2
  3
    8
```

Teaching Tips

Suggested Time: 6–8 min; Difficulty: Easy

- **Name each value:** `t.branches` is a list; `t.branches[i]` is a `Tree`. Have students state the type of each intermediate expression before taking another attribute.
- **Leaves have no branches:** A leaf's `branches` is the empty list, so indexing into it raises an `IndexError`. Contrast with linked lists, where the empty case `()` is not a `Link` at all.
- **Reading print:** Depth is shown by indentation; students can redraw the tree from the printout.

5. Implement `tree_height`, which takes in a tree and returns its height: the depth of the lowest leaf, where the root has depth 0.

```
def tree_height(t: Tree[int]) -> int:
    """
    >>> tree_height(Tree(1))
    0
    >>> t = Tree(1, [Tree(4), Tree(2, [Tree(3)])])
    >>> tree_height(t)
    2
    """
    if _____________________________________________:

        return _____________________________

    return 1 + ____________________________________________________________
```

```
def tree_height(t: Tree[int]) -> int:
    if is_leaf(t):
        return 0
    return 1 + max([tree_height(b) for b in t.branches])
```

Teaching Tips

Suggested Time: 5 min; Difficulty: Easy

- **Use the lecture framework:** What small choice do we make (which branch?), what is the recursive call for each option, and how do we combine the results? Here, call `tree_height` on every branch and take the **`max`**.
- **Why the base case matters:** **`max`** of an empty list errors, so leaves must be handled separately.

6. Write `all_paths`, which takes a tree and returns a list of all root-to-leaf paths. Each path is a linked list of the labels from the root down to a leaf.

```
def all_paths[T](t: Tree[T]) -> list[LinkedList[T]]:
    """
    >>> t = Tree(1, [Tree(2), Tree(3, [Tree(4), Tree(5)])])
    >>> for p in all_paths(t):
    ...     print(p)
    (1 2)
    (1 3 4)
    (1 3 5)
    >>> all_paths(Tree(7))
    [Link(first=7, rest=())]
    """
    if ____________________________________________:

        return ____________________________________

    return ____________________________________________________________________
```

```
def all_paths[T](t: Tree[T]) -> list[LinkedList[T]]:
    if is_leaf(t):
        return [Link(t.label)]
    return [Link(t.label, p) for b in t.branches for p in all_paths(b)]
```

Teaching Tips

Suggested Time: 10 min; Difficulty: Medium

- **Read the return type:** `all_paths` returns a Python list whose elements are linked lists. Ask what `all_paths(b)` means for a branch `b`: every path from that branch's root down to a leaf.
- **Same pattern as partitions:** Add the current label to the front of each path from a branch with `Link(t.label, p)`, as the lecture did with `Link(m, p)`.
- **Nested comprehension:** Read it as two loops: for each branch, for each path in that branch.
- **Common bugs:** Both come from mixing up a list of paths with a single path.
  - Returning `Link(t.label)` in the base case (a path, not a list of paths) gives `TypeError: 'Link' object is not iterable` when the caller loops over the result.
  - Writing `[Link(t.label, all_paths(b)) for b in t.branches]` puts a Python list in `rest`, so printing a path raises `AssertionError: Not a valid linked list`.

7. Write `count_paths`, which takes a tree `t` of integers and an integer `total`, and returns the number of root-to-leaf paths whose labels sum to `total`.

```
def count_paths(t: Tree[int], total: int) -> int:
    """
    >>> t = Tree(1, [Tree(3), Tree(3, [Tree(4), Tree(0)]), Tree(4,
        [Tree(1)])])
    >>> count_paths(t, 4)
    2
    >>> count_paths(t, 8)
    1
    >>> count_paths(t, 6)
    1
    >>> count_paths(t, 5)
    0
    """
    if ____________________________________________:

        return ____________________________________

    return ____________________________________________________________________
```

```
def count_paths(t: Tree[int], total: int) -> int:
    if is_leaf(t):
        return 1 if t.label == total else 0
    return sum([count_paths(b, total - t.label) for b in t.branches])
```

Teaching Tips

Suggested Time: 10 min; Difficulty: Medium

- **Shrink the problem:** Once we use the root label, each branch only needs to produce a sum of `total - t.label`.
- **Base case:** A leaf is a complete path; it counts exactly when its label equals the remaining total.
- **Combine:** Add the counts from all branches, as in `count_leaves` from lecture.
- **Stopping early:** A path must end at a leaf. `count_paths(t, 5)` is the only doctest that catches a solution that returns 1 as soon as the remaining total matches an internal node's label: the path 1, 4 sums to 5 but continues to the leaf 1. Ask students to trace it.